INCH-POUND

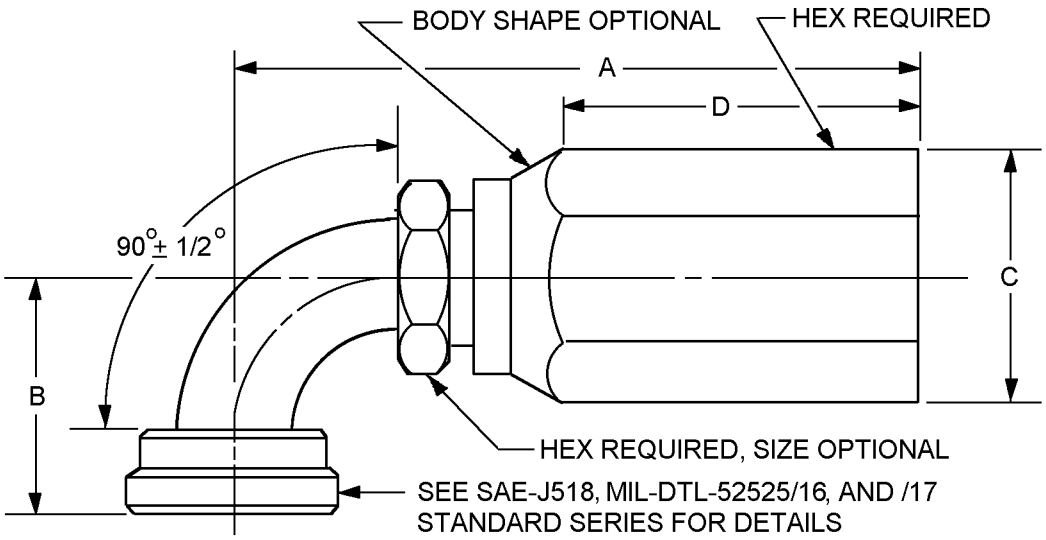
MIL-DTL-52525/12D
21 April 2011
SUPERSEDING
MIL-DTL-52525/12C
20 November 2008

DETAIL SPECIFICATION SHEET

FITTINGS, HOSE, SCREW ON TO SPLIT-FLANGE 90° BENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-52525.



Part or Identifying Number (PIN) hose - flange	Hose ID ref inches (mm)	Flange size inches (mm)	A max inches (mm)	B inches (mm) ±0.12 (3.1)
M52525/12-8-8	.500 (12.70)	.500 (12.70)	3.65 (92.71)	1.63 (41.40)
M52525/12-12-12	.750 (19.05)	.750 (19.05)	4.40 (111.76)	2.13 (54.10)
M52525/12-12-16	.750 (19.05)	1.000 (25.40)	4.40 (111.76)	2.13 (54.10)
M52525/12-16-16	1.000 (25.40)	1.000 (25.40)	5.14 (130.56)	2.38 (60.45)
M52525/12-16-20	1.000 (25.40)	1.250 (31.75)	5.14 (130.56)	2.38 (60.45)
M52525/12-20-20	1.250 (31.75)	1.250 (31.75)	6.33 (160.78)	2.62 (66.55)
M52525/12-20-24	1.250 (31.75)	1.500 (38.10)	6.33 (160.78)	2.69 (68.33)
M52525/12-24-24	1.500 (38.10)	1.500 (38.10)	7.01 (178.05)	3.12 (79.25)
M52525/12-32-32	2.000 (50.80)	2.000 (50.80)	8.62 (218.95)	4.50 (114.30)

FIGURE 1. Fitting, screw-on to 4-bolt split-flange, 90° bent tube.

AMSC N/A FSC 4730

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PIN	C max inches (mm)	D min inches (mm)
M52525/12-8-8	1.38 (35.05)	1.27 (32.26)
M52525/12-12-12	1.75 (44.45)	1.36 (34.54)
M52525/12-12-16	1.75 (44.45)	1.36 (34.54)
M52525/12-16-16	2.06 (52.32)	1.61 (40.89)
M52525/12-16-20	2.06 (52.32)	1.61 (40.89)
M52525/12-20-20	2.63 (66.80)	1.80 (45.72)
M52525/12-20-24	2.63 (66.80)	1.80 (45.72)
M52525/12-24-24	2.94 (74.68)	2.08 (52.83)
M52525/12-32-32	3.50 (88.90)	2.44 (61.98)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. All dimensions not shown shall be in accordance with SAE-J516 and SAE-J518-1 as applicable.
4. SAE-J516 type, for use with SAE-J517, type 100R2 hose coupled to port using SAE-J518 split flange type

FIGURE 1. Fitting, screw-on to 4-bolt split-flange, 90° bent tube - Continued.

REQUIREMENTS:

Fittings shall be accordance with figure 1 and tables I and II.

The fittings described herein are for use with MIL-DTL-52471/5, type 100R2, double wire braided hose and hose assemblies in accordance with MIL-DTL-52525/2.

If fittings are to be used on an oxygen hose, see MIL-DTL-52525.

Materials and finishes shall be in accordance with MIL-DTL-52525 and in table I. All finishes shall be capable of withstanding 96 hours minimum of salt spray.

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TABLE I. Material and finish identification codes. 1/ 2/

PIN code material/plating finish	Material	Plating finish
A	Aluminum alloy 6061 or 7075	Anodize in accordance with MIL-A-8625, type II.
AN		Anodize in accordance with MIL-A-8625, type II and NAVAIR trivalent chromium pretreatment (TCP) in accordance with MIL-DTL-81706, type II, class 1A.
BN		Bare aluminum with NAVAIR TCP in accordance with MIL-DTL-81706, type II, class 1A.
C	Steel 3/	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2.
CN		Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2 and NAVAIR TCP in accordance with MIL-DTL-81706, type II, class 1A.
YC	Chrome-molybdenum steel alloy 4130	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2.
DN		Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2 and NAVAIR TCP in accordance with MIL-DTL-81706, type II, class 1A.
F	Steel 3/	NAVAIR TCP in accordance with MIL-DTL-81706, type II, class 1A.
FN	Chrome-molybdenum steel alloy 4130	
G	Steel 3/	Zinc plating with colorless passivate in accordance with ASTM B633, type VI, Fe/Zn 25.
YG	Chrome-molybdenum steel alloy 4130	
H	Steel 3/	Zinc phosphate finish in accordance MIL-DTL-16232, type Z, class1. 4/
YH	Chrome-molybdenum steel alloy 4130	
J	Steel 3/	Zinc plating with chromate conversion in accordance with ASTM B633; type II or III, Fe/Zn 5, or ASTM B695, type II, class 5.
YJ	Chrome-molybdenum steel alloy 4130	
K	Nickel-copper alloy UNS N04400	No additional finish.
M	High-chromium nickel alloy UNS N06690	No additional finish.
N	Steel 3/	Zinc aluminum in accordance with ASTM F1136, grade 3, NC.
YN	Chrome-molybdenum steel alloy 4130	
P	Steel 3/	Zinc plating with colorless passivate in accordance with ASTM B633, type VI, Fe/Zn 5.
YP	Chrome-molybdenum steel alloy 4130	
S	Corrosion resistant steel	No additional finish. Passivation in accordance with SAE-AMS2700, type 6 or 7.
T	Titanium	Anodize in accordance with SAE-AMS2488, type 2.
V	Steel 3/	Zinc-nickel in accordance with SAE-AMS2417, type 1.
W	Steel 3/	Any zinc plating above.
YW	Chrome-molybdenum steel alloy 4130	
Z	Steel 3/	Any zinc plating above with NAVAIR TCP in accordance with MIL-DTL-81706, type II, class 1A.
YZ	Chrome-molybdenum steel alloy 4130	

1/ All materials and finishes shall be in accordance with MIL-DTL-52525.
2/ Embitterment test need not be run.
3/ All steels except 4130.
4/ Zinc phosphate finish is hexavalent chromium free.

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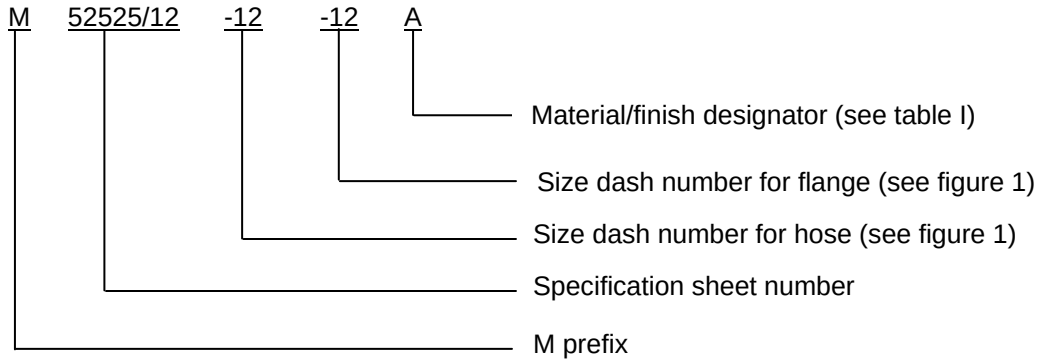
Maximum operating pressure shall be as specified in table II.

TABLE II. Maximum operating pressures of fittings. 1/ 2/

SAE dash size	Hose ID inches (mm)	Steel, CRES, titanium, nickel-copper alloy , and high-chromium nickel		Aluminum	
		psi	MPa	psi	MPa
-8	.500 (12.70)	3500	24.1	2500	17
-12	.750 (19.05)	2250	15.5	1650	11
-16	1.000 (25.40)	2000	13.8	1000	7
-20	1.250 (31.75)	1625	11.2	850	6
-24	1.500 (38.10)	1250	8.6	750	5
-32	2.000 (50.80)	1125	7.8	600	2

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.

PIN: The PIN consists of the letter “M” the specification sheet number, a dash, a number for hose size, a dash, a number for tube size, a dash, a number for flange size, and a material finish designator.



PIN example: M52525/12-12-12A, describes a 90° split flange for a .750 inch (19.05 mm) hose, with a .750 inch (19.05 mm) flange, aluminum with an anodized finish.

To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when the other materials and finishes specified in this document cannot meet performance requirements.

Color identification. Color identification shall be in accordance with SAE-AS4841.

Marking shall include the manufacturer’s name or trademark, size, and hose identifier R2. Location of marking is optional.

Harmonization to SAE-J516 cross reference PIN in table III is for reference only. SAE style fittings may not have the same material/finish as required by MIL-DTL-52525 and legacy with MIL-DTL-52471.

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TABLE III. Harmonization to SAE-J516 cross reference PIN. 1/

PIN	SAE PIN	SAE J518-2 PIN Flange head only
M52525/12-8-8	SAE J516 8-8 391454	SAE J518-2-8-0562
M52525/12-12-12	SAE J516 12-12 391454	SAE J518-2-12-0562
M52525/12-12-16	SAE J516 12-16 391454	SAE J518-2-16-0562
M52525/12-16-16	SAE J516 16-16 391454	SAE J518-2-16-0562
M52525/12-16-20	SAE J516 16-20 391454	SAE J518-2-20-0562
M52525/12-20-20	SAE J516 20-20 391454	SAE J518-2-20-0562
M52525/12-20-24	SAE J516 20-24 391454	SAE J518-2-24-0562
M52525/12-24-24	SAE J516 24-24 391454	SAE J518-2-24-0562
M52525/12-32-32	SAE J516 32-32 391454	SAE J518-2-32-0562

1/ Material and finish designators are omitted from PIN's see table I and SAE-J846.

Notes:

1. For design feature purposes, this standard takes precedence over the procurement document referenced herein. Referenced documents shall be of the issue in effect on the date of invitation for bid.
2. In the event of a conflict between the text of this standard and the referenced cited herein, the text of this standard shall take precedence.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-52525, this document references the following:

MIL-A-8625	MIL-DTL-81706	SAE-AMS2488
MIL-DTL-16232	ASTM B633	SAE-AMS2700
MIL-DTL-52471	ASTM B695	SAE-AS4841
MIL-DTL-52471/5	ASTM F1136	SAE-J516
MIL-DTL-52525/2	SAE-AMS-C-81562	SAE-J517
MIL-DTL-52525/16	SAE-AMS-QQ-P-416	SAE-J518
MIL-DTL-52525/17	SAE-AMS2417	SAE-J846

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CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - SH
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2011-052)

Review activities:
Army - CR4
Navy - AS, CG, MC, SA, YD

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